

5 Monte Carlo Results

5.1 Instrument Relevance and Point Estimation

Table 1 establishes that the four values of κ produce clearly ordered excluded-instrument relevance settings, using the first-stage F -statistic and partial R^2 as descriptive diagnostics.

Both diagnostics fall strongly as κ decreases. For $n = 500$, the median F -statistic falls from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$ and 1.22 at $\kappa = 0.10$; the corresponding end-points for $n = 1000$ are 212.82 and 2.04. Partial R^2 is comparatively stable across sample sizes, whereas F increases with n . The design therefore varies the instruments' explanatory contribution while sample size affects the amount of information available. These diagnostics describe relative relevance within the experiment; they are not quantile-specific weak-IV thresholds.

Table 2 is the central finite-sample performance table. It reports Bias, MAE, RMSE, Coverage, and the median accepted-set measure for $n = 1000$, $\tau \in \{0.10, 0.50, 0.90\}$, all four κ values, and Oracle-GMM, Full-GMM, and DML-IVQR-BC.

As instrument relevance declines, MAE and RMSE generally increase for all three estimators and across the reported quantiles. At $\tau = 0.50$, moving from $\kappa = 1$ to $\kappa = 0.10$ raises RMSE from 0.072 to 0.945 for Oracle, from 0.100 to 1.018 for Full, and from 0.104 to 0.991 for DML. Oracle deterioration shows that the phenomenon is not generated solely by high-dimensional nuisance estimation: knowing the relevant controls does not replace instrument

Table 1. Instrument-strength calibration

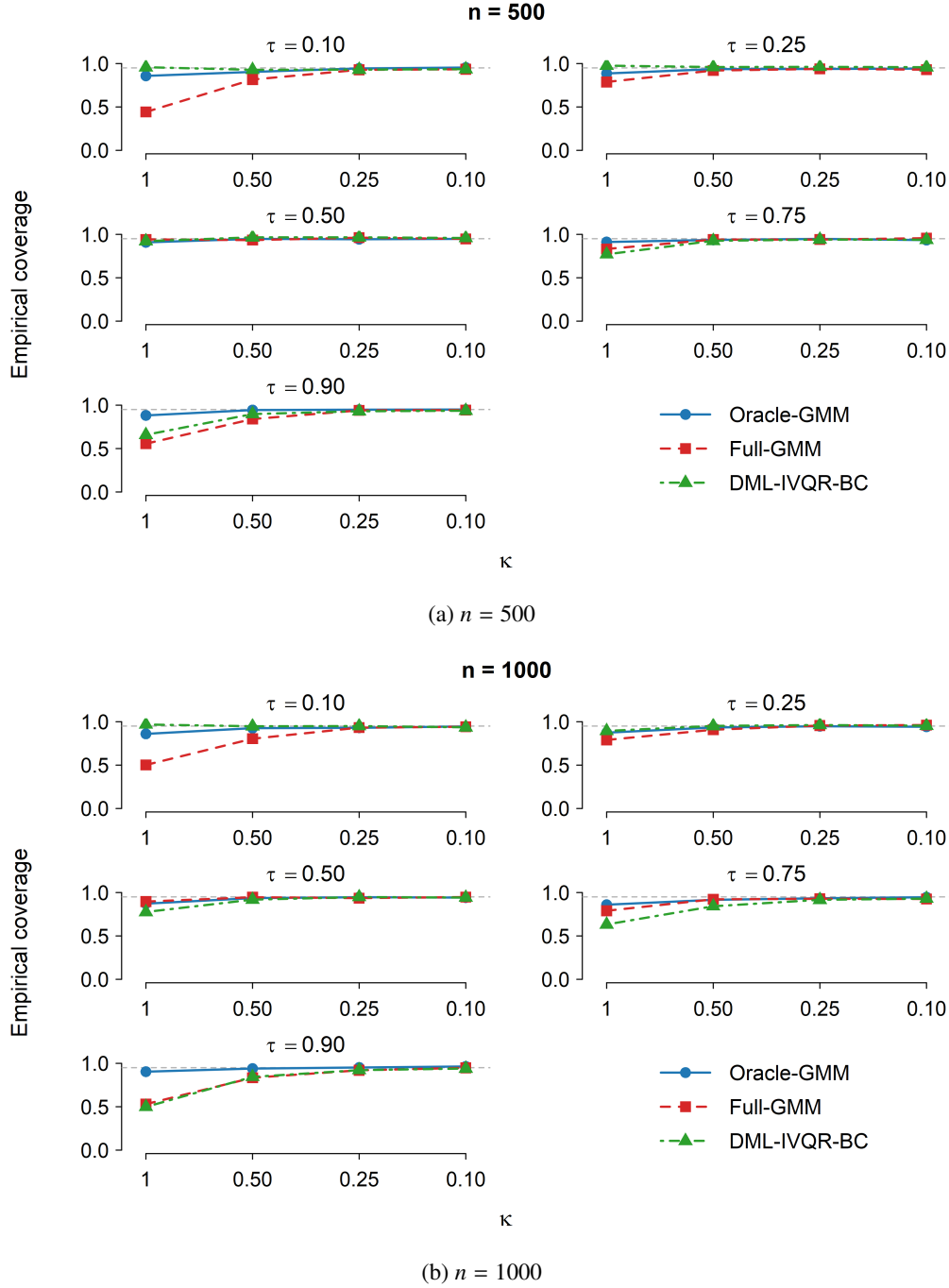
n	κ	Median F	F : p10–p90	Median partial R^2	Partial R^2 : p10–p90
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

Table 2. Main finite-sample performance at $n = 1000$

τ	κ	Estimator	Bias	MAE	RMSE	Coverage	Median accepted-set measure
0.10	1.00	Oracle-GMM	-0.054	0.093	0.121	0.860	0.400
		Full-GMM	-0.329	0.337	0.387	0.502	0.800
		DML-IVQR-BC	-0.005	0.117	0.150	0.968	0.600
	0.50	Oracle-GMM	-0.101	0.194	0.287	0.926	1.000
		Full-GMM	-0.407	0.465	0.630	0.804	1.900
		DML-IVQR-BC	-0.159	0.306	0.482	0.948	2.225
	0.25	Oracle-GMM	-0.384	0.566	0.934	0.930	3.450
		Full-GMM	-0.773	0.914	1.242	0.932	3.500
		DML-IVQR-BC	-0.681	0.844	1.229	0.950	3.750
	0.10	Oracle-GMM	-0.747	0.975	1.376	0.946	4.000
		Full-GMM	-1.074	1.251	1.582	0.942	3.900
		DML-IVQR-BC	-0.932	1.146	1.514	0.936	4.000
0.50	1.00	Oracle-GMM	0.007	0.047	0.072	0.872	0.200
		Full-GMM	0.007	0.072	0.100	0.896	0.400
		DML-IVQR-BC	0.070	0.079	0.104	0.776	0.300
	0.50	Oracle-GMM	0.015	0.093	0.134	0.936	0.500
		Full-GMM	0.014	0.151	0.199	0.946	0.800
		DML-IVQR-BC	0.059	0.119	0.160	0.918	0.600
	0.25	Oracle-GMM	0.045	0.281	0.431	0.946	1.900
		Full-GMM	0.029	0.372	0.500	0.936	2.200
		DML-IVQR-BC	-0.003	0.306	0.448	0.948	2.025
	0.10	Oracle-GMM	-0.044	0.735	0.945	0.942	3.900
		Full-GMM	-0.012	0.840	1.018	0.948	3.800
		DML-IVQR-BC	-0.195	0.779	0.991	0.942	3.900
0.90	1.00	Oracle-GMM	0.072	0.110	0.148	0.904	0.500
		Full-GMM	0.387	0.397	0.465	0.530	1.000
		DML-IVQR-BC	0.183	0.186	0.215	0.498	0.400
	0.50	Oracle-GMM	0.211	0.295	0.540	0.940	1.575
		Full-GMM	0.584	0.629	0.871	0.834	2.450
		DML-IVQR-BC	0.226	0.260	0.396	0.844	1.100
	0.25	Oracle-GMM	0.491	0.658	1.038	0.952	3.650
		Full-GMM	0.901	1.058	1.389	0.920	3.600
		DML-IVQR-BC	0.417	0.580	0.927	0.922	3.350
	0.10	Oracle-GMM	0.653	0.948	1.350	0.964	4.000
		Full-GMM	1.050	1.236	1.583	0.948	3.900
		DML-IVQR-BC	0.519	0.831	1.191	0.940	4.000

Note: Each cell is based on 500 Monte Carlo replications. Bias follows the frozen convention ($\text{Bias} = \alpha_0 - \hat{\alpha}$). Coverage is evaluated directly at the true structural quantile effect. The accepted-set measure is the grid-based measure within the prespecified primary computational domain ($A_0 = [-1, 3]$); it is not an unrestricted confidence-region length.

**Figure 1.** Empirical coverage across instrument-strength designs.

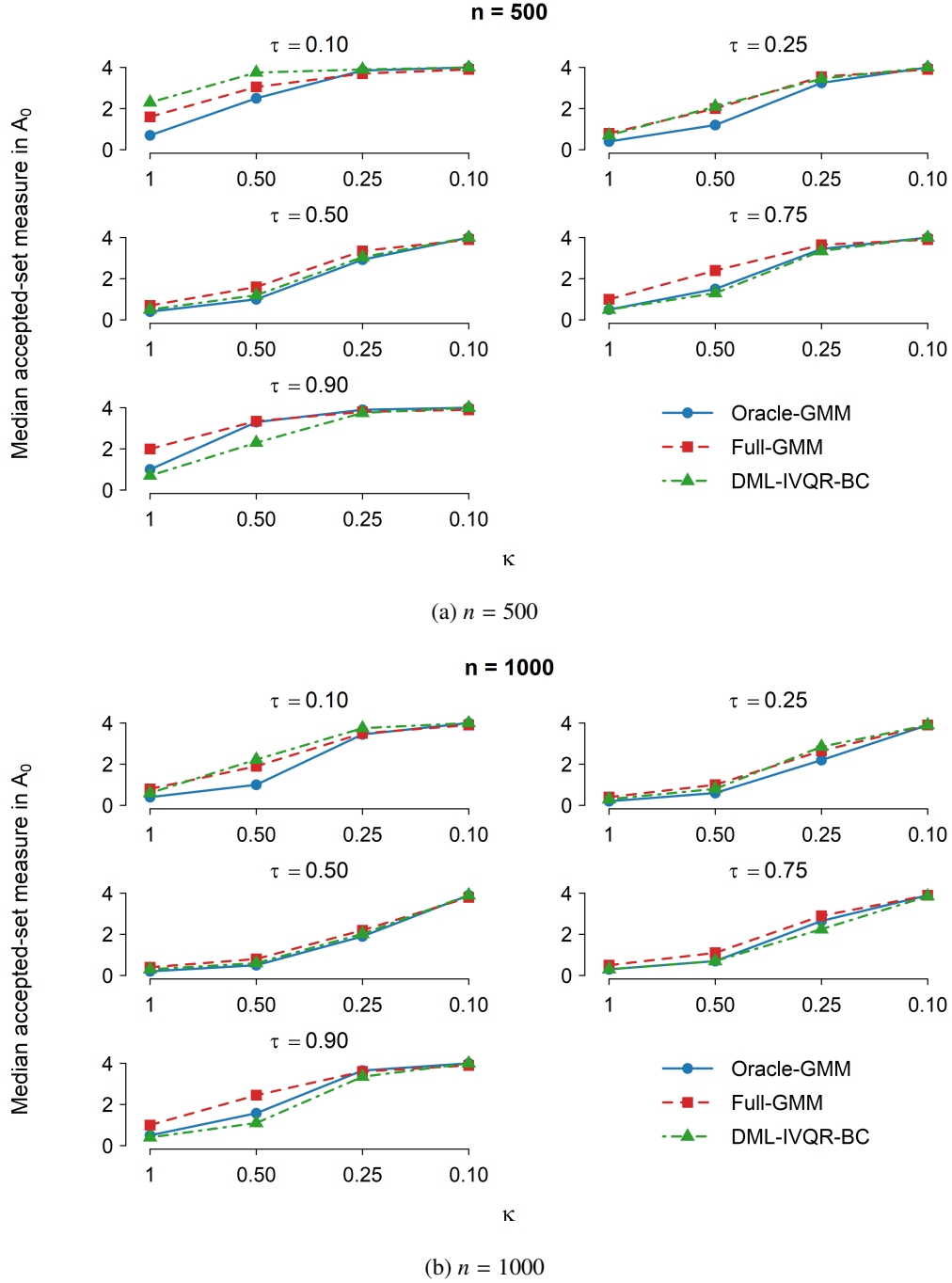


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$.

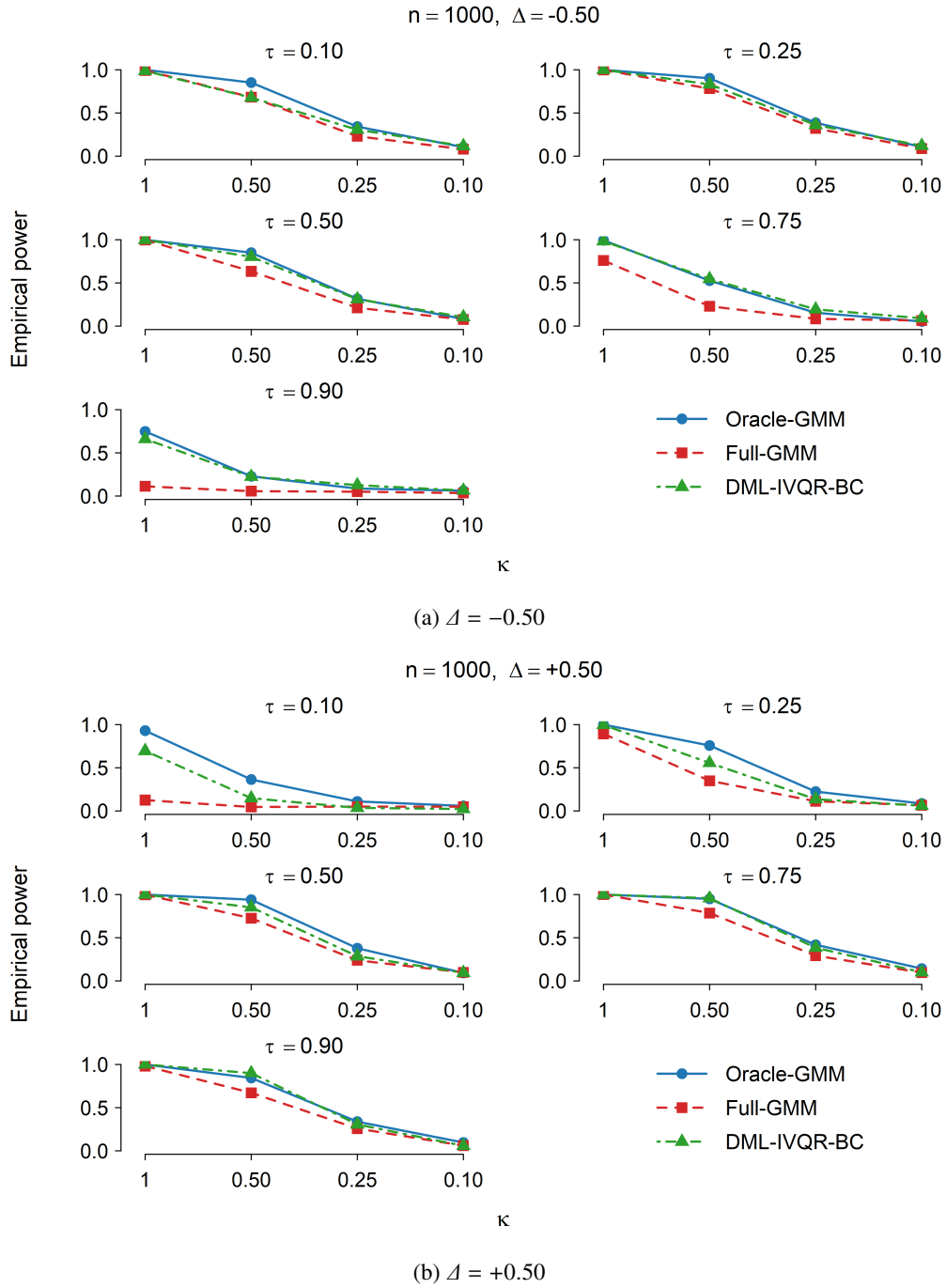


Figure 3. Rejection probability under moderate false alternatives, $n = 1000$.